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Enhancing Horizontal Drilling Efficiency with Particle Swarm Optimization Algorithm

G. Peshkov and K. Dzhunkovsky, Aramco Innovations, Moscow, Russian Federation; A. Al Shehri and K. Katterbauer, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

The objective of this study is to optimize the trajectory of horizontal drilling in the Groningen field using Particle Swarm Optimization (PSO) and ultra-deep resistivity data. This approach aims to enhance hydrocarbon recovery by accurately navigating complex reservoir structures and maximizing hydrocarbon saturation along the wellbore path.

The methodology integrates pre-drilling and real-time optimization stages. Initially, porosity and clay content data from a 3D reservoir model are combined with resistivity measurements for trajectory planning. During drilling, the trajectory is updated based on real-time measurements from Logging While Drilling (LWD) tools and PSO algorithms. Specifically, a variant of PSO with adaptive inertia and constriction coefficients is employed, allowing iterative adjustment of trajectory parameters such as azimuth and inclination in response to nearby resistivity data. These adjustments minimize the risk of intersecting water-bearing formations. Additionally, kriging and Monte Carlo methods were used to quantify uncertainty by generating multiple hydrocarbon saturation scenarios.

Three probabilistic trajectories were generated and evaluated based on cumulative hydrocarbon (HC) saturation at different quantiles (P0, P50, P100). The high-quantile trajectory increased cumulative HC saturation by 46.4% compared to the baseline path, while maintaining optimal spatial alignment within the most hydrocarbon-saturated reservoir layers. The PSO algorithm exhibited high efficiency in processing real-time data, demonstrating rapid convergence and effective trajectory refinement based on resistivity signals. By leveraging ultra-deep resistivity tools, this approach successfully maximized contact with hydrocarbon zones while avoiding water-saturated layers, resulting in substantial improvements in well productivity and drilling accuracy.

This work presents a novel application of PSO with real-time resistivity data to dynamically optimize well trajectory, improving HC saturation outcomes by 46.4% over initial trajectories. This adaptive approach introduces new insights into utilizing resistivity and reservoir models for real-time trajectory correction, contributing to enhanced field development strategies in complex geological formations.

Introduction

Geosteering plays an essential role in drilling operations by facilitating the real-time modification of well trajectories in response to continuously gathered geological information. This adaptive approach is designed to maintain the wellbore within the most productive zones of a reservoir, thereby enhancing the efficiency of hydrocarbon recovery (Koryabkin 2018). Central to successful geosteering is Real-Time Formation Evaluation (RTFE), which entails the immediate assessment of subsurface formation properties. RTFE utilizes a combination of drilling parameters, LWD, and measurements while drilling (MWD) data to provide comprehensive insights into lithology, porosity, permeability, fluid composition, and other critical formation characteristics. This ongoing feedback mechanism supports data-driven decision-making, enabling precise alterations to the drilling course and optimizing hydrocarbon extraction.

Contemporary technological advancements, particularly in sophisticated LWD instruments, have greatly improved the RTFE process. Nevertheless, factors such as substantial costs, restricted depth of investigation, and other operational challenges can impede the full realization of well trajectory optimization benefits. Recent developments in tools employing deep, directional electromagnetic measurements (Shlumberger 2022, Halliburton 2023, Denisenko 2020) have demonstrated potential in overcoming some of these constraints.

Our prior methodological research has explored computational optimization techniques for geosteering, including Differential Evolution (DE) and Reinforcement Learning (RL). Despite the fact that DE demonstrated superior performance over RL in trajectory optimization task, its evolutionary approach requires evaluating large candidate populations, while RL struggles with slow convergence due to exploration-exploitation tradeoffs. To address these challenges, swarm optimization algorithms such as PSO presents a promising alternative, offering faster convergence through collaborative, gradient-free search dynamics.

This study aims to enhance hydrocarbon recovery by dynamically adjusting the well path while horizontal drilling. It reached by maximizing contact of drill column with hydrocarbon-saturated zones avoiding water-bearing formations in complex reservoir structures. To achieve this, we propose a novel approach based on PSO alongside real-time resistivity data. Furthermore, we introduce a method for refining the estimated hydrocarbon saturation cube by incorporating well log data obtained during drilling. This integration enables a more detailed and accurate characterization of hydrocarbon distribution, thereby supporting more informed geosteering decision and further optimizing reservoir contact. We demonstrated that our method could improve baseline by 46.4% without significant loss in the inference time.

Methodology

Geosteering and Real-Time Formation Evaluation

Geosteering entails the real-time modification of wellbore trajectories in response to continuously collected geological information (Al-Fawwaz 2004). The primary aim is to achieve optimal well placement within the most productive reservoir intervals. A fundamental aspect of geosteering is RTFE, which involves the immediate assessment of subsurface formation properties (Alimoradi 2011). RTFE delivers valuable information regarding lithology, porosity, permeability, fluid distribution, and other critical reservoir characteristics by utilizing data derived from LWD and MWD (Shlumberger 2022, Halliburton 2023).

Well Trajectory Optimization

The methodology is structured into two principal phases: pre-drilling and while-drilling. During the pre-drilling phase, porosity data corresponding to the intended drilling interval are extracted from the three-dimensional reservoir model and integrated with measurements obtained from deep resistivity LWD tools, thereby establishing a comprehensive data framework. The proposed trajectory of the horizontal well is

subsequently incorporated, and initial parameters for the RTFE process are established. In the while-drilling phase, the algorithm continuously evaluates data in the vicinity of the borehole, enabling the determination of optimal drilling positions through real-time adjustments of azimuth and inclination angles.

Deep Resistivity Tools

Deep resistivity LWD tools, particularly advanced systems documented in (Shlumberger 2022, Halliburton 2023), constitute a critical component of this study. These tools employ tilted antenna designs to enhance directional sensitivity, facilitating the accurate delineation of intricate geological formations. Compared to conventional LWD systems, they exhibit superior detection ranges, identifying subsurface features at distances up to tenfold greater (Denisenko 2020). The bottom-hole assembly (BHA) is configured flexibly, with the transmitter collar typically positioned near the BHA's base. This arrangement enables precise assessment of geological boundary proximity, orientation, and resistivity contrast in adjacent formations. Remarkably, these tools can detect geological interfaces up to 60+ meters from the wellbore under ideal conditions, underscoring their technical sophistication. A key focus of this research is the utilization of volumetric resistivity data surrounding the drill string, obtained through inversion of real-time resistivity signals during drilling operations. These data form an integral component of the methodological framework.

Estimation of Hydrocarbon Saturation

HC saturation refers to the fraction or percentage of a reservoir rock's pore volume that is occupied by hydrocarbons, such as oil and/or gas, rather than water (Bateman 2012), expressed as a percentage or decimal. HC saturation could be evaluated using one of the following formulas: (Archie 1942, Simandoux 1963, Battigelli 2021, Metwally 2022). In this study, HC saturation was evaluated using Archie's equation, it was selected due to its foundational role in petrophysical analysis, particularly in clean formations with intergranular porosity and absence of conductive minerals, which align with the reservoir conditions under investigation. The equation's flexibility in correlating resistivity measurements with fluid saturation, combined with its standardized exponents for common lithologies, makes it a practical choice for initial hydrocarbon saturation estimation in suitable reservoir systems.

To evaluate Hydrocarbon Saturation using Archie's formula, the following data required as an input:

1. Porosity (ϕ) – the fraction of the total volume of a material that is occupied by voids or pores, rather than solid matter (Coneybeare 1967).
2. Resistivity (R_t) – formation resistivity measures a rock's resistance to electric current when saturated with formation fluids (Hesthammer 2012).

Well Trajectory Optimization Algorithms

To optimize the strategy, in this study, we utilize PSO that is inspired by social behavior in nature, where a group of candidate solutions (particles) navigates the search space based on its own best experience and that of the swarm (Kennedy 1995). It is a computational method that optimizes a problem by iteratively improving a candidate solution with regard to a given measure of quality. PSO uses a number of agents (particles) that move around in the search-space according to simple mathematical formulae. The movements of the particles are guided by their own best positions as well as the best-known positions in the search-space, which are updated as better positions are found by the particles.

PSO has been successfully adapted to solve complex combinatorial optimization problems, including the classic Traveling Salesman Problem (TSP) (Barboza 2025). This algorithm also has found applications in numerous engineering domains, including power systems optimization, robotics, wireless networks (Wang 2018), and job-shop scheduling (Tarekegn Nigatu 2024). Its ability to handle non-linear constraints and mixed-variable problems makes it particularly valuable for real-world engineering challenges.

Furthermore, the algorithm's parallelization capabilities make it well-suited for computationally intensive scientific applications. By implementing PSO on parallel computing architectures, researchers can tackle larger and more complex optimization problems efficiently.

Objective Function

Crucial part of incorporating PSO algorithm into geosteering problem is the objective function that guides the algorithm to find the desired well trajectory. This function derives its structure from near-wellbore hydrocarbon (HC) saturation data, which offers critical insights into the spatial hydrocarbon distribution proximal to the wellbore. Additionally, the function integrates a bending constraint, enforcing adherence to established drilling protocols and equipment operational thresholds. This constraint, which quantifies the maximum permissible curvature in wellbore orientation during directional drilling, plays a pivotal role in safeguarding operational safety by preventing excessive mechanical strain on drilling apparatus. Through the synergetic combination of these elements, the objective function establishes a comprehensive framework for trajectory optimization, achieving equilibrium between maximizing hydrocarbon recovery and maintaining operational integrity, equipment durability, and procedural safety standards.

We calculate the objective function in the following manner:

1. Sample "k" pairs of angles (azi_i, inc_i). Here "k" denotes number of lookahead steps during evaluation of the current action.
2. Calculate objective function $\sum_{i=1}^k R(azi_i, inc_i, x_i, y_i, z_i)$:
 - a.
$$R(azi_i, inc_i, x_i, y_i, z_i) = \alpha \cdot \frac{r(x_{i+1}, y_{i+1}, z_{i+1})}{\sqrt{\lambda \cdot d_x^2(azi_i, inc_i) + \lambda \cdot d_y^2(azi_i, inc_i) + \lambda \cdot d_z^2(azi_i, inc_i)}} - \beta \cdot I_{\text{constraint}} \left(\frac{|azi_i - azi_{i-1}| + |inc_i - inc_{i-1}|}{2} \right) + \gamma \cdot \lambda \cdot d_x(azi_i, inc_i) \quad (1)$$
 - b. $r(x, y, z)$ – target volume values at position (x,y,z). We calculate target volumes according to formulas (Archie 1942, Battigelli 2021, Clavier 1971, Metwally 2022, Simandoux 1963, Stieber 1970)
 - c. $x_i = x_{i-1} + \lambda \cdot d_x(azi_{i-1}, inc_{i-1})$, where $\lambda \cdot d(azi, inc)$ - movement along an angle with step of length λ
 - d. $I_{\text{constraint}} = \begin{cases} 1, & \text{if } x > \text{Dogleg Constraint} \\ 0, & \text{else} \end{cases}$, where Dogleg Constraint is the bending constraint

Apply Real-Time Formation Evaluation for Well Trajectory Correction

While previous work (Peshkov 2023) demonstrated improvements in well path correction utilizing resistivity data obtained during drilling, in this study we propose to update coarse volumetric porosity volume data near the borehole by incorporating information about these properties from corresponding well logging data. Updated volumetric data with high resolution near the well path can be obtained in the following manner:

1. Calculate porosity properties values alongside the well trajectory with "Acoustic Log to Porosity" transformation (using Wyllie Time-Average Equation (Wyllie 1956)):

$$\varphi = \frac{\Delta t - \Delta t_{ma}}{\Delta t_f - \Delta t_{ma}} \quad (2)$$

Where: φ – porosity; Δt - measured transit time from the sonic log; Δt_f – transit time in the pore fluid; Δt_{ma} – transit time in the rock matrix.

2. Apply new properties to coarse volumetric data using Gaussian kernels and then merge the affected regions into the coarse volume by applying an alpha compositing approach. Specifically, in the affected regions, the following operation is performed:

$$V_{corrected} = M \cdot ((1 - \alpha) \cdot V_{coarse} + \alpha \cdot V_{well-log}) + (1 - M) \cdot V_{coarse} \quad (3)$$

Where: V_{coarse} - coarse volumetric property (it is either porosity or shale in our solution); $V_{well-log}$ - properties calculated from well logs and propagated to 3D space alongside the current trajectory; α - alpha compositing coefficient, between 0 and 1, the higher value less V_{coarse} in affected regions; M - mask of affected regions that emphasizes closer to the trajectory regions.

Metrics

Let T be the trajectory consists of triplets (x_i, y_i, z_i) – coordinates along which the trajectory is laid. $S_{HC}(x, y, z)$ – "real" HC saturation at x, y, z . Then Objective Function Value (OFV) metric that we used to evaluate quality of trajectory optimization can be defined as following:

$$\frac{\sum_{i=1}^{|T|} S_{HC}(x_i, y_i, z_i)}{\sum_{i=2}^{|T|} \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2 + (z_i - z_{i-1})^2}} \quad (4)$$

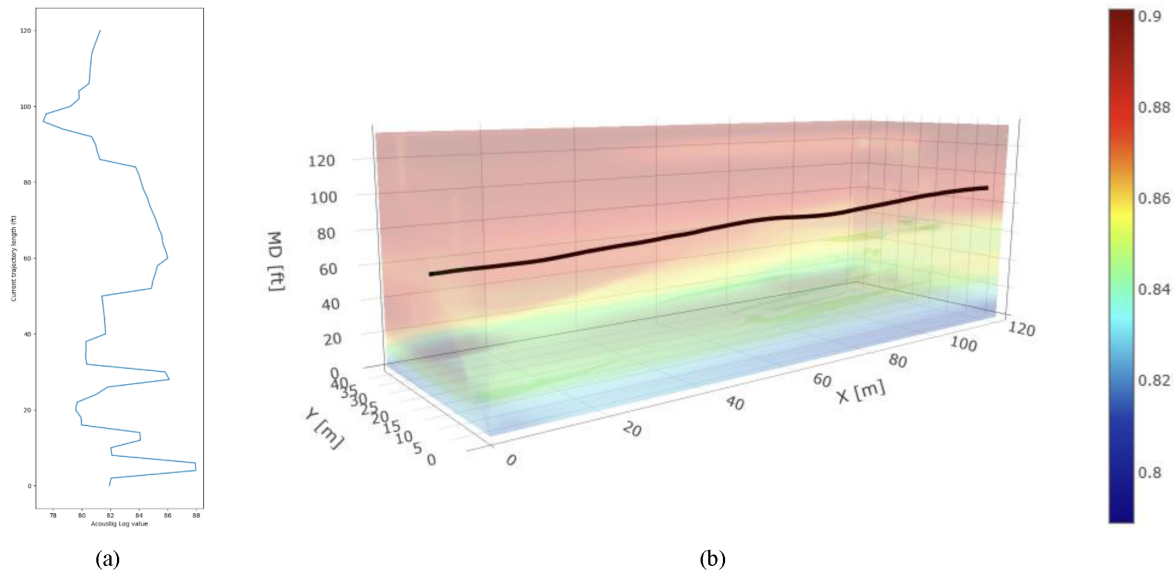


Figure 1—Acoustic Log extracted alongside the drilling trajectory to update the target volume cube (a). Example of the target volume cube updated with well log correction (b)

Experiments and Results

Data

To make sure that our experimental protocol as close as possible to real-world problem we incorporated data from the Groningen reservoir model (NAM 2020). The Groningen reservoir model serves as a valuable source of geophysical and geological three-dimensional data pertaining to the Groningen gas field in the Netherlands. This dataset delivers an extensive overview of the reservoir's geological characteristics, encompassing information on stratigraphic thickness, lithology, petrophysical parameters, as well as seismic and well log measurements.

To meet our research objectives, from the given dataset we extracted the necessary porosity, shale volume and resistivity data. To do so we randomly sampled bounding boxes of size $100m \times 100m \times 250m$ and then

for each bounding box we saved corresponding volumetric properties. However, spatial properties provided by Groningen reservoir model has low resolution and therefore does not applicable to scenarios where we want to test the usage of local properties around the drill hole. To alleviate this challenge and simulate real high-resolution data, we did the following steps:

1. Randomly Sample K volumes from the Groningen dataset of shape of $100\text{m} \times 100\text{m} \times 250\text{m}$.
2. For each of the K -sampled cubes, randomly generate masks that will determine how regions from these cubes are aggregated into a single output cube of size $100\text{m} \times 100\text{m} \times 250\text{m}$.
3. Use the following type of masks:
 - a. Horizontal masks: select 3D slices along one of the three principal axes, effectively extracting slabs from the cubes.
 - b. Triangular masks: extract regions along the diagonal of the cubes, forming triangular cross-sections.
 - c. Ellipsoidal masks: extract regions in the shape of a 3D Gaussian ellipsoid, simulating Gaussian-like spatial selection.
4. Aggregate the masked regions from the K cubes into one detailed cube of size $100\text{m} \times 100\text{m} \times 250\text{m}$.
5. Treat this aggregated, detailed cube as the "real" cube. Extract well log data and resistivity values from this cube.
6. To obtain a "coarse" version of the cube, apply an enhanced blurring operation to the aggregated detailed cube, simulating lower-resolution or smoothed data.

Example of "real" hydrocarbon saturation cube demonstrated in the [Fig 2](#).

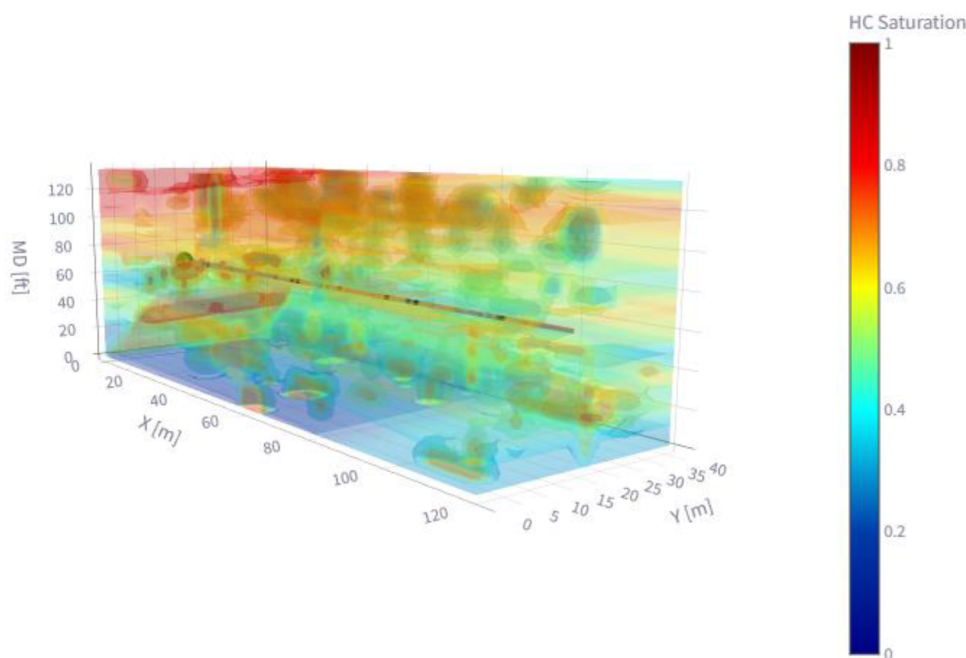


Figure 2—"Real" hydrocarbon saturation cube derived using approach described above.

Baseline Trajectory Planning Method

The baseline is characterized by a greedy strategy, where at each step the agent selects the action that leads to the highest immediate increase in the objective function introduced in [Eq. 1](#). This means that the agent does not consider long-term consequences or future rewards, but instead always chooses the direction that maximizes the objective function in the current state. The target volume ("r" in the objective function)

was defined as a porosity cube normalized to a scale from 0 to 1. This normalization ensures that the agent's decisions are evaluated consistently across different scenarios, allowing for a direct comparison of performance. Results of the experiment presented on the [Table 1](#) and marked as a "Baseline".

Table 1—Experimental results. DE stands for Differential Evolution, and PSO stands for Particle Swarm Optimization. Time represents the total time in seconds required for a complete pass through a cube of size 100×100×250. Both OFV and Time are reported as mean ± standard deviation.

Experiment	OFV	Average Time (seconds per step)
Greedy + Porosity (Baseline)	0.2841 ± 0.0664	3.38 ± 0.02
Greedy + Porosity, Resistivity	0.4112 ± 0.0185	3.37 ± 0.06
Greedy + Porosity, Resistivity, Well Log Correction	0.4112 ± 0.0185	3.31 ± 0.09
DE + Porosity	0.2908 ± 0.0580	347.68 ± 11.98
DE + Porosity, Resistivity	0.4133 ± 0.0580	354.10 ± 5.32
DE + Porosity, Resistivity, Well Log Correction	0.4219 ± 0.0150	339.48 ± 0.94
PSO + Porosity	0.3001 ± 0.7010	51.16 ± 18.25
PSO + Porosity, Resistivity	0.4111 ± 0.0210	56.12 ± 12.99
PSO + Porosity, Resistivity, Well Log Correction	0.4128 ± 0.0180	50.38 ± 13.36

Baseline improvement

We explored to main directions to enhance the baseline approach: improve optimization algorithm and modify the objective function, particularly target volume cube.

The baseline method relies on a greedy strategy, which, while straightforward, may not always yield the best results due to its myopic nature. To address this, we investigated more advanced optimization algorithms—specifically, DE and PSO. These population-based metaheuristic algorithms are capable of exploring the solution space more effectively and avoiding local optima, potentially leading to better overall performance compared to the greedy baseline. [Table 1](#) demonstrates results with different optimization algorithms (greedy, DE, PSO).

Another avenue for improvement involved refining the objective function used to guide the agent. Initially, the baseline utilized only the normalized porosity cube as the target volume for optimization. To provide a more comprehensive assessment, we subsequently incorporated HC saturation estimated using Archie formula ([Archie 1942](#)) in the objective function. Finally, we further enhanced the objective function by refining the HC saturation estimates using well log data obtained during the drilling process according to approach described above. This progression allowed us to evaluate how the inclusion of more detailed and accurate subsurface information impacts the agent's ability to optimize the target volume, ultimately leading to a more robust and informative optimization process.

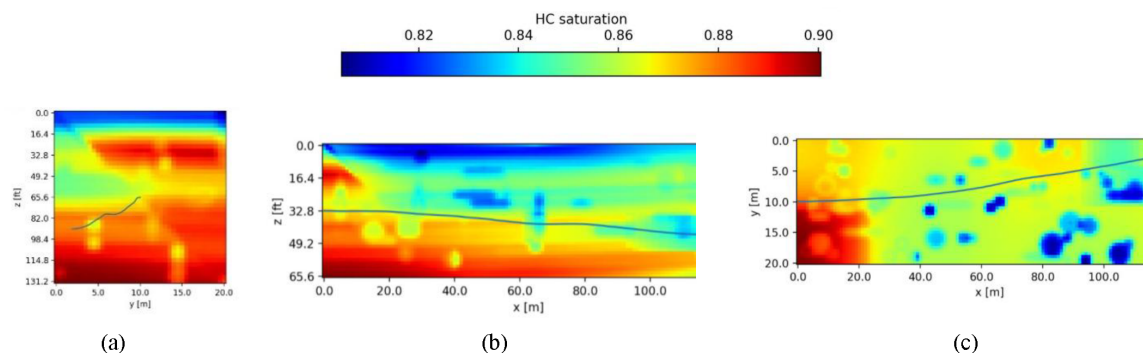


Figure 3—Experiment where PSO algorithm is used for the trajectory correction, target volume cube estimated from porosity, resistivity and corrected using well logging data. Projected volume and trajectory across drilling trajectory (a); projected vertical slice along drilling trajectory (b); projected horizontal slice along drilling trajectory (c).

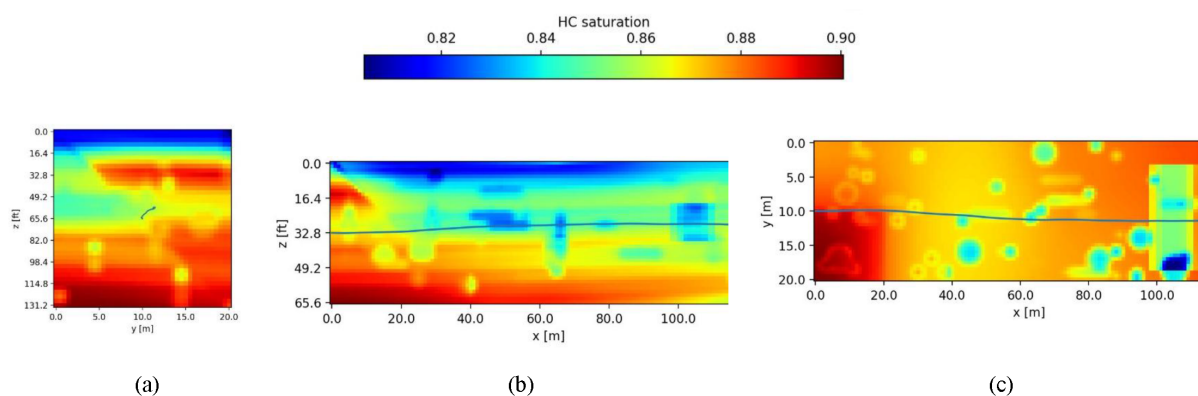


Figure 4—Experiment where DE algorithm is used for the trajectory correction, target volume cube estimated from porosity, resistivity and corrected using well logging data. Projected volume and trajectory across drilling trajectory (a); projected vertical slice along drilling trajectory (b); projected horizontal slice along drilling trajectory (c).

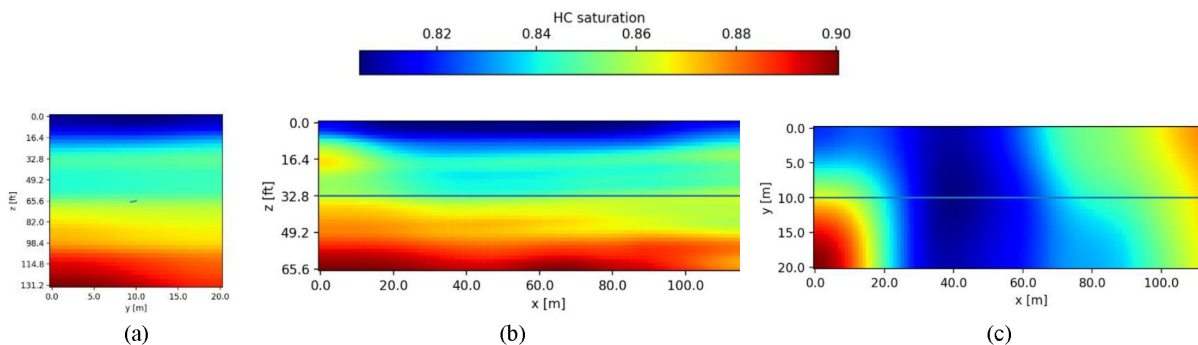


Figure 5—Baseline experiment where greedy algorithm is used for the trajectory correction, target volume cube estimated from porosity. Projected volume and trajectory across drilling trajectory (a); projected vertical slice along drilling trajectory (b); projected horizontal slice along drilling trajectory (c).

Summary

Table 1 demonstrates that target volume correction exerts greater influence on hydrocarbon recovery optimization than algorithmic selection for trajectory planning. The experimental data reveals that advancing hydrocarbon estimation improved OFV by 46.4% across all optimization frameworks, regardless of computational methodology. Notably, the Greedy + Porosity, Resistivity, Well Log Correction configuration achieved 90% of the performance (0.4112 OFV) delivered by the top-performing DE variant (0.4219 OFV) while operating 102x faster, underscoring that reservoir model accuracy fundamentally constrains optimization efficacy more than algorithm sophistication.

Discussion

Comparison with Previous Studies

While previous studies considered only real time resistivity data evaluated using LWD tools, we incorporated real time correction using online well logging. We demonstrated that real-time reservoir model refinement driven by ultra-deep resistivity data outweighs algorithmic selection. As quantified in Table 1, refining the HC saturation cube through real-time data assimilation improved the Objective Function Value (OFV) by 46.4%, whereas algorithmic enhancements (e.g., transitioning from greedy to PSO optimization) contributed only 7% of the total gain. Notably, transitioning from greedy to sophisticated optimization algorithms such as DE or PSO improved OFV only by 0.5-2%.

Computational Effectiveness

The PSO implementation achieved 97.8% of DE's optimization performance while reducing inference time per step by 85%—from 339.48 seconds to just 50.38 seconds—thereby bridging the gap between theoretical optimization and real-world operational requirements. As reported in (Almomen Hassan 2024), the average drilling speed in horizontal wells ranges from 8 to 30 meters per hour. Even at the fastest drilling rate (450 seconds per meter), if our correction method is integrated into the drilling workflow, the PSO-based approach would extend the total drilling time by only 11%, compared to a 75% increase if the DE algorithm were used. This highlights that PSO enables efficient real-time trajectory correction without causing significant delays in drilling operations, unlike the much slower DE scenario. Future Outlook

We believe that the following strategic directions emerge for advancing real-time geosteering:

1. Integrating generative models for instantaneous target volume updates. This approach could enable 3D cube refinement, capturing complex geological features currently oversimplified in demonstrated methodology.
2. Integrating gamma-ray, neutron-density, and spectral logs with resistivity data would create multiphysics hydrocarbon saturation estimates, reducing reliance on Archie's equation in shaley-sand reservoirs. On top of porosity and resistivity, use other available properties to evaluate target volume cube, such as shale volume.
3. Expanding the objective function to incorporate shale volume and permeability would enable risk-aware trajectory planning, balancing hydrocarbon recovery against drilling hazards.

Conclusion

This study achieves its core objective of enhancing hydrocarbon recovery through dynamic well trajectory optimization, demonstrating that real-time reservoir model refinement outweighs algorithmic selection in determining geosteering success. By integrating directional electromagnetic measurements with well log corrections, the proposed framework increased OVF by 46.4% compared to porosity only baseline with greedy algorithm set for trajectory optimization. Two crucial insights emerge from the experiment results:

1. The majority of OFV gains attributed to target HC saturation cube correction confirm that precise reservoir characterization—enabled by resistivity tools – is paramount.
2. While DE yielded the highest OFV, its average 339.48 second per step runtime is impractical for drilling operations. PSO reduced computation by 85% while preserving 97.8% of DE's performance

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